Arithmetic Positivity and Absolute Hodge Classes: Why Hodge-Riemann Positivity Does Not Strengthen Deligne's Absoluteness

Lukas Geiger[1]

[1] *Independent Researcher, Bernau, Germany*

(Dated: May 6, 2026)

We introduce the concept of *arithmetic positivity* for the Hodge Conjecture: a rational (p, p)-class $\alpha \in H^{2p}(X, \mathbb{Q})$ on a smooth projective variety $X/\mathbb{C}$ is called arithmetically positive if it simultaneously satisfies Hodge-Riemann positivity under all polarizations, Deligne's absoluteness under all automorphisms of $\mathbb{C}$, and Lefschetz compatibility. We prove that algebraic classes are arithmetically positive and record limited functoriality tests: finite pullback is controlled on the pullback ample subcone and becomes a full AP statement only under an explicit ample-cone comparison hypothesis; Künneth products are treated for primitive classes and product-generated ample cones; finite pushforward remains conjectural.

Our **main result is a No-Go theorem**: the arithmetically positive cone coincides exactly with Deligne's absolute Hodge classes ($\mathrm{AP}^p(X) = \mathrm{AbsHodge}^p(X)$), because the Hodge-Riemann bilinear relations automatically guarantee the positivity conditions for any class of type (p, p). This is a well-known consequence of the Hodge-Riemann relations; the contribution of this paper is to make the saturation boundary explicit and to delineate what Hodge-Riemann-based positivity methods cannot achieve. The natural attempt to strengthen Deligne's absoluteness through Hodge-Riemann positivity therefore produces no new constraints, and the Arithmetic Positivity Conjecture reduces to Deligne's question (1982): "Are all absolute Hodge classes algebraic?" Furthermore, the obstructions (O1) and (O3) in the No-Go formulation are vacuous for rational (p, p)-classes; only (O2), failure of absoluteness, is effective. We introduce the Galois-Hodge-Riemann spectrum as a visualization of the absoluteness test, illustrate the theoretical predictions numerically across selected test configurations, and discuss controlled test-family directions for finding genuinely stronger conditions beyond the Hodge-Riemann relations.

This paper is an expository contribution documenting a structural limitation; it does not contain new results beyond the observation that AP = AbsHodge.

ORCID: https://orcid.org/0009-0005-7296-1534

AI tools (Claude by Anthropic) were used for mathematical formalization, literature verification, and editorial refinement. All mathematical content was developed, verified, and approved by the author.

I. INTRODUCTION

A. The Hodge Conjecture

Let X be a smooth projective variety of dimension n over $\mathbb{C}$. The Hodge decomposition

$$H^k(X, \mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X) \tag{1}$$

expresses the k-th singular cohomology as a direct sum of Dolbeault cohomology groups. A class $\alpha \in H^{2p}(X, \mathbb{Q})$ is called a *rational Hodge class* if its image in $H^{2p}(X, \mathbb{C})$ lies entirely in $H^{p,p}(X)$. Every algebraic cycle Z of codimension p defines a rational Hodge class $\mathrm{cl}(Z) \in H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X)$ via the cycle class map.

Conjecture I.1 (Hodge, 1950 [1])**.** *Every rational Hodge class on a smooth projective variety over $\mathbb{C}$ is a $\mathbb{Q}$-linear combination of classes of algebraic cycles.*

Despite over seven decades of effort, the conjecture remains open in general. It is known to hold for:

- Divisors ($p = 1$): by the Lefschetz $(1, 1)$-theorem.
- Abelian varieties of dimension ≤ 5: by work of Hazama [9] and Moonen [10].
- Products of elliptic curves and certain Fermat hypersurfaces.
- Uniruled varieties (for degree-2 classes).

B. Why construction fails

All major approaches to the Hodge Conjecture share a common strategy: *construct* the algebraic cycle representing a given Hodge class. This paper argues that this strategy is fundamentally misguided, and proposes an alternative: prove that non-algebraic Hodge classes are *impossible* via a positivity obstruction.

The structural failures of existing approaches are:

1. Direct algebraization. Given a rational (p, p)-class, attempt to construct a subvariety whose fundamental class matches it. This fails because there is no canonical projection from Hodge cohomology to algebraic cycles, and the construction is not functorial under deformation.

2. Variations of Hodge structure. Use families X_t and track how Hodge classes vary. This fails because

Hodge classes can “jump” – the condition of being type (p, p) is not open in the Zariski topology – and monodromy destroys naive stability arguments [2].

3. Motivic methods. Replace varieties by motives and hope that Hodge classes are motivic. This fails because the category of motives is not fully constructed: Grothendieck’s standard conjectures [13] remain open, and the relationship between Hodge classes and absolute Hodge classes is unresolved [5].

4. Reduction mod p. Reduce to characteristic p, invoke the Tate conjecture, and lift back. This fails because the bridge Hodge $\Rightarrow$ Tate $\Rightarrow$ Hodge is not closed: Frobenius invariance does not imply Hodge type, and lifting algebraic cycles is not unique [6].

As we show below, the positivity alternative proposed in this paper also encounters a fundamental limitation: the natural positivity conditions reduce to Deligne’s absoluteness. Nevertheless, the analysis delineates a precise boundary for this class of approaches.

C. The positivity alternative

We observe that major breakthroughs in adjacent conjectures follow a different pattern:

Conjecture	Key positivity property
Weil conjectures	Non-negativity of Frobenius eigenvalues
BSD (rank 1)	Non-negativity of Néron–Tate height
Tate (abelian var.)	Positivity via Faltings’ isogeny theorem

In each case, a positivity property of the relevant arithmetic object plays a structural role – though the proof techniques differ substantially (Deligne’s proof of the Weil conjectures uses ℓ-adic cohomology and the Lefschetz fixed-point theorem; Gross–Zagier explicitly constructs a Heegner point and computes its height). The Hodge Conjecture lacks: *a positivity structure that forces algebraicity without constructing cycles.*

This paper introduces such a structure – *arithmetic positivity* – combining Hodge-Riemann positivity under all polarizations with Deligne’s absoluteness under all automorphisms of $\mathbb{C}$. The central finding, however, is a *No-Go result*: the Hodge-Riemann bilinear relations automatically guarantee the positivity conditions for any class of type (p, p), so the arithmetically positive cone coincides with Deligne’s absolute Hodge classes: $\mathrm{AP}^p(X) = \mathrm{AbsHodge}^p(X)$. The automatic saturation of Hodge-Riemann positivity for (p, p)-classes is well known to specialists (cf. Voisin [2], Chapter 6; Griffiths–Harris [3], Chapter 0–1); the purpose of this paper is to make the saturation boundary explicit and systematic within a unified positivity framework, and to clarify that no Hodge-Riemann-based strengthening of Deligne’s absoluteness is possible. The Arithmetic Positivity Conjecture reduces to Deligne’s question: “Are all absolute Hodge classes algebraic?”

We regard this boundary result as informative: it delineates a precise limit for positivity-based approaches to the Hodge Conjecture and identifies directions for genuinely stronger conditions beyond Hodge-Riemann.

D. Outline

Section II reviews the Hodge-Riemann bilinear form and absolute Hodge classes. Section III introduces arithmetic positivity. Section IV proves that algebraic classes are arithmetically positive. Section V establishes functoriality. Section VI formulates the No-Go theorem. Section VII proves the equivalence for abelian varieties. Section VIII identifies the precise obstruction. Section IX discusses the remaining gap and the relationship to Deligne’s program.

This paper is self-contained; all main results depend only on definitions and proofs within this document. Connections to the broader FST programme are cited for context but are not logically required.

II. BACKGROUND

A. Hodge-Riemann bilinear form

Let X be a smooth projective variety of dimension n, equipped with a Kähler form ω defining an ample class $L = [\omega] \in H^{1,1}(X) \cap H^2(X, \mathbb{Z})$. The *Lefschetz operator* $\Lambda_L : H^k(X) \to H^{k+2}(X)$ is defined by $\Lambda_L(\alpha) = L \cup \alpha$. We use the convention $\Lambda_L(\alpha) = L \cup \alpha$; some authors reserve Λ for the dual Lefschetz operator, while denoting the cup-product operation simply by L or $L\wedge$ (Voisin [2]; Griffiths–Harris [3]).

Theorem II.1 (Hard Lefschetz)**.** *For $k \leq n$, the map $\Lambda_L^{n-k} : H^k(X, \mathbb{Q}) \to H^{2n-k}(X, \mathbb{Q})$ is an isomorphism.*

A class $\alpha \in H^k(X, \mathbb{Q})$ with $k \leq n$ is *primitive* (with respect to L) if $\Lambda_L^{n-k+1}(\alpha) = 0$. The *Hodge-Riemann bilinear form* on the primitive cohomology $P^k(X, \mathbb{Q}) \subset H^k(X, \mathbb{Q})$ is:

$$Q_L(\alpha, \beta) = \int_X \alpha \cup \beta \cup L^{n-k} . \tag{2}$$

Theorem II.2 (Hodge-Riemann bilinear relations)**.** *On the primitive subspace $P^{p,q}(X) = P^{p+q}(X) \cap H^{p,q}(X)$ with $p + q = k$, the Hermitian form*

$$h_L(\alpha, \beta) = i^{p-q} (-1)^{k(k-1)/2} Q_L(\alpha, \bar{\beta}) \tag{3}$$

is positive definite.

The Hodge-Riemann relations are the fundamental positivity result in Hodge theory. They provide definiteness *relative to a fixed polarization L*. A central theme of this paper is that algebraicity requires positivity not just for one L, but for *all* admissible polarizations and operations.

B. Absolute Hodge classes

Deligne [4] introduced a refinement of Hodge classes. An automorphism $\sigma \in \mathrm{Aut}(\mathbb{C})$ sends a smooth projective variety $X/\mathbb{C}$ to the conjugate variety X^σ (same equations, coefficients twisted by σ). The comparison isomorphism provides:

$$H^{2p}_{\mathrm{dR}}(X/\mathbb{C}) \xrightarrow{\sim} H^{2p}_{\mathrm{dR}}(X^\sigma/\mathbb{C}), \tag{4}$$

mapping the de Rham class of α on X to a class α^σ on X^σ.

Definition II.3 (Deligne)**.** A rational Hodge class $\alpha \in H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X)$ is *absolute* (or *absolutely Hodge*) if for every $\sigma \in \mathrm{Aut}(\mathbb{C})$, the conjugate class α^σ is again a Hodge class on X^σ.

Theorem II.4 (Deligne [4])**.** *1. Every algebraic class is absolute.*

2. On abelian varieties, every Hodge class is absolute.

Deligne's result shows that absoluteness is a *necessary* condition for algebraicity. The question is whether it is sufficient. Our concept of arithmetic positivity *attempts* to strengthen absoluteness by adding Hodge-Riemann positivity under all conjugations; as shown in Section IX, this attempt does not succeed (the additional conditions are automatic), but the analysis is informative.

III. ARITHMETIC POSITIVITY

We now introduce the central concept.

A. Definition

Let X be a smooth projective variety of dimension n over $\mathbb{C}$, and let $\mathrm{Amp}(X)$ denote the ample cone of X.

Definition III.1 (Arithmetic Positivity)**.** A rational Hodge class $\alpha \in H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X)$ is *arithmetically positive* if it satisfies the following three conditions simultaneously:

(AP1) Universal Hodge-Riemann positivity. For every ample class $L \in \mathrm{Amp}(X)$, the primitive component α_0 of α (with respect to L) satisfies the *signed* Hodge-Riemann inequality:

$$(-1)^p\, Q_L(\alpha_0, \alpha_0) \geq 0\,, \tag{5}$$

where Q_L is the Hodge-Riemann form (2). The sign $(-1)^p$ is essential: the Hodge-Riemann bilinear relations (Theorem II.2) give $(-1)^p Q_L > 0$ on primitive (p,p)-classes, not $Q_L > 0$.

(AP2) Absolute stability. For every $\sigma \in \mathrm{Aut}(\mathbb{C})$:

(a) α^σ is a Hodge class on X^σ (i.e., α is absolute).

(b) The conjugated class α^σ satisfies (AP1) on X^σ with respect to every ample class $L^\sigma \in \mathrm{Amp}(X^\sigma)$.

(AP3) Lefschetz compatibility. For all $0 \leq j \leq n-2p$ and all ample L, let $\gamma_j = \Lambda_L^j \alpha \in H^{2(p+j)}(X)$ and let $\gamma_{j,0}$ denote its primitive component with respect to L in $H^{2(p+j)}(X)$. Then:

$$(-1)^{p+j}\, Q_L(\gamma_{j,0},\, \gamma_{j,0}) \geq 0\,. \tag{6}$$

The set of arithmetically positive classes in $H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X)$ is denoted $\mathrm{AP}^p(X)$.

The full AP condition is *a priori* stronger than Deligne's absoluteness alone; we will show in Section 8 that the two notions in fact coincide.

Remark III.2. The conditions (AP1), (AP2b), and (AP3) are *not* independent of (AP2a). As shown in Proposition IX.1 below, the Hodge-Riemann bilinear relations automatically guarantee (AP1), (AP2b), and (AP3) for any class of type (p,p). The only substantive condition is therefore (AP2a): Deligne's absoluteness. In particular, $\mathrm{AP}^p(X) = \mathrm{AbsHodge}^p(X)$ (Remark IX.2). The definition retains all three conditions for expository clarity: they make explicit the *type* of positivity that algebraic classes enjoy, even though the Hodge-Riemann relations render them automatic for (p,p)-classes. Finding genuinely stronger conditions – beyond the Hodge-Riemann relations – that *do* constrain beyond absoluteness remains an open problem (see Section IX).

B. The Arithmetic Positivity Conjecture

Conjecture III.3 (Arithmetic Positivity Conjecture)**.** *Let X be a smooth projective variety over $\mathbb{C}$. Then:*

$$\mathrm{AP}^p(X) = \mathrm{cl}(\mathrm{CH}^p(X)_{\mathbb{Q}})\ , \tag{7}$$

where $\mathrm{CH}^p(X)_{\mathbb{Q}}$ is the Chow group of codimension-p cycles with rational coefficients, and cl *is the cycle class map.*

Proposition III.4. *The following relations hold between the APC and the HC:*

1. *The Hodge Conjecture implies the Arithmetic Positivity Conjecture.*
2. *The "hard direction" of the APC ($\mathrm{AP}^p(X) \subseteq \mathrm{cl}(\mathrm{CH}^p(X)_{\mathbb{Q}})$) together with the statement that all rational Hodge classes are arithmetically positive ($H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X) \subseteq \mathrm{AP}^p(X)$) implies the HC.*
3. *Unconditionally:* $\mathrm{cl}(\mathrm{CH}^p(X)_{\mathbb{Q}}) \subseteq \mathrm{AP}^p(X)$ *(Theorem IV.1 below).*

Proof. (1) If the HC holds, then every Hodge class is algebraic, and algebraic classes satisfy (AP1)–(AP3) (Theorem IV.1 below). Thus $\mathrm{AP}^p(X) \subseteq H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X) = \mathrm{cl}(\mathrm{CH}^p(X)_{\mathbb{Q}})$, and the reverse inclusion holds by Theorem IV.1, giving $\mathrm{AP}^p(X) = \mathrm{cl}(\mathrm{CH}^p(X)_{\mathbb{Q}})$.

(2) This requires two separate assumptions. *First*, suppose that the hard direction of the APC holds: $\mathrm{AP}^p(X) \subseteq \mathrm{cl}(\mathrm{CH}^p(X)_{\mathbb{Q}})$, i.e., every arithmetically positive class is algebraic. *Second*, suppose that every rational Hodge class is arithmetically positive: $H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X) \subseteq \mathrm{AP}^p(X)$. Under both assumptions, every rational Hodge class α lies in $\mathrm{AP}^p(X)$ (by the second assumption) and hence in $\mathrm{cl}(\mathrm{CH}^p(X)_{\mathbb{Q}})$ (by the first assumption), so α is algebraic. This yields the HC.

Note that neither assumption alone suffices: the APC ($\mathrm{AP}^p = \mathrm{alg}^p$) shows that AP classes are algebraic, but without knowing that all Hodge classes are AP, one cannot conclude that all Hodge classes are algebraic.

(3) See Theorem IV.1. □

Remark III.5. The APC alone does not directly imply the HC. The APC asserts $\mathrm{AP}^p = \mathrm{alg}^p$, while the HC asserts $\mathrm{Hodge}^p = \mathrm{alg}^p$. Since $\mathrm{AP}^p \subseteq \mathrm{Hodge}^p$ (arithmetic positivity imposes additional conditions beyond being a Hodge class), the APC gives $\mathrm{alg}^p \subseteq \mathrm{Hodge}^p$ (trivial) but not $\mathrm{Hodge}^p \subseteq \mathrm{alg}^p$ without the additional knowledge that all Hodge classes are arithmetically positive. The non-trivial content thus splits into two parts: (i) show that non-AP Hodge classes do not exist (i.e., every Hodge class is AP), and (ii) show that AP implies algebraic (the hard direction of APC).

The non-trivial content of the APC is the "hard direction": $\mathrm{AP}^p(X) \subseteq \mathrm{cl}(\mathrm{CH}^p(X)_{\mathbb{Q}})$. This says that arithmetic positivity *forces* algebraicity.

IV. ALGEBRAIC CLASSES ARE ARITHMETICALLY POSITIVE

Theorem IV.1 (Easy Direction). *Let X be a smooth projective variety over $\mathbb{C}$. Every class in the image of the cycle class map is arithmetically positive:*

$$\mathrm{cl}(\mathrm{CH}^p(X)_{\mathbb{Q}}) \subseteq \mathrm{AP}^p(X)\,. \tag{8}$$

Proof. Let $\alpha = \mathrm{cl}(Z)$ for some algebraic cycle Z of codimension p with rational coefficients. We verify (AP1)–(AP3).

(AP1): Let α_0 be the L-primitive component of $\alpha = \mathrm{cl}(Z)$. Since α is algebraic, it lies in $H^{p,p}(X) \cap H^{2p}(X,\mathbb{Q})$, and its primitive component α_0 lies in $P^{p,p}(X) \cap H^{2p}(X,\mathbb{Q})$. By the Hodge-Riemann bilinear relations (Theorem II.2), the Hermitian form h_L is positive definite on $P^{p,p}(X)$. For a rational (p,p)-class, $\bar{\alpha}_0 = \alpha_0$ (complex conjugation acts as the identity on classes of type (p,p) with rational coefficients), $i^{p-p} = 1$, and $(-1)^{k(k-1)/2} = (-1)^{p(2p-1)}$ with $k = 2p$. Thus $h_L(\alpha_0,\alpha_0) = (-1)^{p(2p-1)}\, Q_L(\alpha_0,\alpha_0)$. Since $(-1)^{p(2p-1)} = (-1)^{2p^2-p} = (-1)^p$ (as $(-1)^{2p^2} = 1$), the Hodge-Riemann relations give:

$$(-1)^p\, Q_L(\alpha_0,\alpha_0) > 0 \tag{9}$$

if $\alpha_0 \neq 0$. For $\alpha_0 = 0$ (i.e., when α has no primitive component), the inequality holds trivially. This argument uses only the Hodge-Riemann relations on $P^{p,p}(X)$ and does not require α_0 itself to be the class of an effective cycle.

(AP2): Part (a) is Deligne's theorem (Theorem II.4): algebraic classes are absolute. Part (b) follows because σ sends algebraic cycles on X to algebraic cycles on X^σ (algebraicity is a property of polynomial equations, which transform covariantly under σ), and algebraic classes on X^σ satisfy (AP1) by the argument above.

(AP3): The Hard Lefschetz theorem is compatible with the cycle class map: if $\alpha = \mathrm{cl}(Z)$, then $\Lambda_L^j\alpha = \mathrm{cl}(Z \cdot L^j)$, where $Z \cdot L^j$ is the intersection with j copies of a hyperplane section. This is again an algebraic cycle of codimension $p+j$. Its L-primitive component $\gamma_{j,0}$ lies in $P^{p+j,\,p+j}(X)$. Since $\gamma_{j,0}$ is the primitive part of an algebraic class, the Hodge-Riemann relations give $(-1)^{p+j}Q_L(\gamma_{j,0},\gamma_{j,0}) \geq 0$, by the same argument as in (AP1). □

V. FUNCTORIALITY OF ARITHMETIC POSITIVITY

For the No-Go approach to work, the arithmetically positive cone must be functorial: it must be preserved by the standard operations of algebraic geometry.

Proposition V.1 (Pullback – finite morphisms). *Let $f : Y \to X$ be a finite morphism of smooth projective varieties. If $\alpha \in \mathrm{AP}^p(X)$, then $f^*\alpha$ satisfies (AP1) for ample classes in the pullback subcone $f^*\mathrm{Amp}(X) \subset \mathrm{Amp}(Y)$, and satisfies (AP2)–(AP3) on that subcone. It is a full element of $\mathrm{AP}^p(Y)$ only under an explicit ample-cone comparison hypothesis, for example when the relevant positivity inequalities extend from $f^*\mathrm{Amp}(X)$ to all of $\mathrm{Amp}(Y)$ by density or by equality of the numerical ample cones.*

Proof. Let $f : Y \to X$ be finite of degree d.

(AP1): Since f is finite, f^*L_X is ample on Y for every ample $L_X \in \mathrm{Amp}(X)$. For the primitive component α_0 of α with respect to L_X, the projection formula gives:

$$\int_Y f^*\alpha_0 \cup f^*\bar{\alpha}_0 \cup (f^*L_X)^{n-2p} = d\int_X \alpha_0 \cup \bar{\alpha}_0 \cup L_X^{n-2p},$$

where $n = \dim X = \dim Y$. Since $d > 0$ and the right side satisfies $(-1)^p Q_{L_X}(\alpha_0,\alpha_0) \geq 0$ by assumption, we obtain $(-1)^p Q_{f^*L_X}(f^*\alpha_0, f^*\alpha_0) \geq 0$ on the pullback ample subcone. For a general ample class $L_Y \in \mathrm{Amp}(Y)$, no extension follows from finiteness alone: the primitive component of $f^*\alpha$ varies with L_Y, and the pullback subcone need not be dense in the ample cone of Y. If an additional ample-cone comparison hypothesis gives density or equality of the relevant cone, then continuity of the Hodge-Riemann form and of the Lefschetz decomposition extends the inequality to all ample classes.

(AP2): Pullback commutes with σ-conjugation: $(f^*\alpha)^\sigma = (f^\sigma)^*(\alpha^\sigma)$, and $f^\sigma : Y^\sigma \to X^\sigma$ is again finite of the same degree. Since $\alpha^\sigma \in \mathrm{AP}^p(X^\sigma)$ by assumption, the result follows from the (AP1) argument applied to f^σ and α^σ.

(AP3): The Lefschetz operator satisfies $\Lambda^j_{f^*L} \circ f^* = f^* \circ \Lambda^j_L$ (compatibility of pullback with cup product). Therefore $\Lambda^j_{f^*L}(f^*\alpha) = f^*(\Lambda^j_L\alpha)$, and the positivity of the primitive component follows from the same degree-d argument applied to $\Lambda^j_L\alpha$. □

Remark V.2. The degree formula $Q_{f^*L_X}(f^*\alpha_0, f^*\alpha_0) = \deg(f) \cdot Q_{L_X}(\alpha_0, \alpha_0)$ used in the proof relies on f being a finite morphism (equivalently, generically finite with $\dim Y = \dim X$). For a generically finite morphism $f : Y \to X$ with $\dim Y = \dim X$ that is not finite, the projection formula still yields the degree relation on a dense open subset, but care is needed regarding the ramification locus; the statement of Proposition V.1 is therefore restricted to finite morphisms.

Remark V.3. For morphisms $f : Y \to X$ with $\dim Y > \dim X$, the pullback f^*L_X is not ample on Y, and the primitive decomposition on Y requires a relative Hodge-Riemann analysis. The functoriality of AP^p under general morphisms remains open.

Remark V.4 (Density caveat). The extension from $f^*\mathrm{Amp}(X)$ to all of $\mathrm{Amp}(Y)$ requires an explicit ample-cone comparison. A useful sufficient condition is that the pullback subcone has dense image in the relevant numerical ample cone, or that the numerical ample cones agree after pullback. This may hold in Picard-number-one situations, but it fails in general. Without such a hypothesis, Proposition V.1 is only a subcone functoriality statement.

Conjecture V.5 (Pushforward – finite morphisms)**.** *Let $f : Y \to X$ be a finite morphism of smooth projective varieties (so $\dim Y = \dim X$). If $\alpha \in \mathrm{AP}^p(Y)$, then $f_*\alpha \in \mathrm{AP}^p(X)$.*

Evidence. The projection formula $f_*f^* = \deg(f) \cdot \mathrm{id}$ on $H^*(X, \mathbb{Q})$ implies that f_* is injective on the image of f^* (up to scalar). For $\alpha \in \mathrm{AP}^p(Y)$ and any ample $L_X \in \mathrm{Amp}(X)$, the class f^*L_X is ample on Y (since f is finite). The Hodge-Riemann form on X is related to that on Y via the adjunction $\int_X f_*\beta \cup \gamma = \int_Y \beta \cup f^*\gamma$. Combined with the degree formula, the sign conditions for $f_*\alpha$ follow from those of α. The absolute stability transfers because f^σ is again finite of the same degree. A complete proof requires verifying that the primitive decomposition on X is compatible with f_* applied to primitive classes on Y; this is not straightforward, since f_* does not preserve primitivity in general. □

Remark V.6. For morphisms $f : Y \to X$ with $\dim Y > \dim X$, the pushforward f_* involves a Gysin-type argument relating the Hodge-Riemann form on Y to that on X via the Grothendieck-Riemann-Roch formula and the relative primitive decomposition. This analysis is not carried out here; the functoriality of AP^p under general pushforward remains open.

Proposition V.7 (Künneth product – primitive classes, product polarizations)**.** *Let $\alpha \in \mathrm{AP}^p(X)$ and $\beta \in \mathrm{AP}^q(Y)$ be primitive with respect to some ample classes L_X and L_Y respectively. Then $\alpha \boxtimes \beta$ satisfies (AP1) for all product polarizations $L = L_X \boxtimes 1 + 1 \boxtimes L_Y$, and satisfies (AP2)–(AP3). In particular, $\alpha \boxtimes \beta \in \mathrm{AP}^{p+q}(X \times Y)$ when product polarizations generate* $\mathrm{Amp}(X \times Y)$ *(e.g., when X or Y has Picard number* 1*); for general product varieties, the extension to all ample classes remains open.*

Remark V.8. The extension to non-primitive classes requires a careful analysis of cross-terms in the Lefschetz decomposition on the product, which we do not carry out here. The result for primitive classes covers the structurally essential case.

We first establish a key lemma on primitivity of exterior products.

Lemma V.9 (Primitivity of exterior products)**.** *Let X and Y be smooth projective varieties of dimensions n and m respectively, with ample classes $L_X \in \mathrm{Amp}(X)$ and $L_Y \in \mathrm{Amp}(Y)$. Set $L = L_X \boxtimes 1 + 1 \boxtimes L_Y \in \mathrm{Amp}(X \times Y)$. If $\alpha \in P^{2p}(X, \mathbb{Q})$ is primitive with respect to L_X and $\beta \in P^{2q}(Y, \mathbb{Q})$ is primitive with respect to L_Y, then $\alpha \boxtimes \beta$ is primitive with respect to L on $X \times Y$.*

Proof. Set $k = p + q$ and $N = n + m = \dim(X \times Y)$. We must show $L^{N-2k+1} \cup (\alpha \boxtimes \beta) = 0$. Since $L = L_X \boxtimes 1 + 1 \boxtimes L_Y$, the binomial expansion gives for any $j \geq 0$:

$$L^j \cup (\alpha \boxtimes \beta) = \sum_{a+b=j} \binom{j}{a} (L_X^a \cup \alpha) \boxtimes (L_Y^b \cup \beta).$$

Each summand vanishes unless both factors are non-zero. Since α is L_X-primitive, $L_X^{n-2p+1} \cup \alpha = 0$, so $L_X^a \cup \alpha = 0$ for $a \geq n-2p+1$. Similarly, $L_Y^b \cup \beta = 0$ for $b \geq m-2q+1$. A summand with $a + b = j$ is non-zero only if $a \leq n - 2p$ and $b \leq m - 2q$, which requires $j \leq (n-2p) + (m-2q) = N - 2k$. Therefore $L^j \cup (\alpha \boxtimes \beta) = 0$ for all $j \geq N - 2k + 1$, which is precisely the primitivity condition. □

Lemma V.10 (Sign compatibility for exterior products)**.** *Under the hypotheses of Lemma V.9, with $k = p + q$:*

$$(-1)^k\, Q_L(\alpha \boxtimes \beta,\, \alpha \boxtimes \beta) = \binom{N-2k}{n-2p} \big[(-1)^p Q_{L_X}(\alpha, \alpha)\big] \cdot \big[(-1)^q Q_{L_Y}(\beta, \beta)\big], \tag{10}$$

where $N = n + m$ and Q_L, Q_{L_X}, Q_{L_Y} are the Hodge-Riemann forms (2) *on $X \times Y$, X, and Y respectively.*

Proof. By Lemma V.9, $\alpha \boxtimes \beta$ is L-primitive, so the definition (2) gives directly:

$$Q_L(\alpha \boxtimes \beta,\, \alpha \boxtimes \beta) = \int_{X\times Y} (\alpha \boxtimes \beta)^2 \cup L^{N-2k}.$$

Expanding L^{N-2k} via the binomial formula as in Lemma V.9, the only non-vanishing term occurs at $a = n-2p$, $b = m-2q$ (forced by $a \le n-2p$, $b \le m-2q$, and $a+b = N-2k = (n-2p)+(m-2q)$). By the Künneth formula for integration:

$$\int_{X\times Y} (\alpha \boxtimes \beta)^2 \cup L^{N-2k} = \binom{N-2k}{n-2p} \left(\int_X \alpha^2 \cup L_X^{n-2p}\right) \cdot \left(\int_Y \beta^2 \cup L_Y^{m-2q}\right).$$

By definition (2), the integrals on X and Y are precisely $Q_{L_X}(\alpha,\alpha)$ and $Q_{L_Y}(\beta,\beta)$. Therefore:

$$Q_L(\alpha \boxtimes \beta,\, \alpha \boxtimes \beta) = \binom{N-2k}{n-2p} Q_{L_X}(\alpha,\alpha) \cdot Q_{L_Y}(\beta,\beta).$$

Multiplying both sides by $(-1)^k = (-1)^{p+q} = (-1)^p \cdot (-1)^q$ yields equation (10). ☐

Proof of Proposition V.7. We verify (AP1)–(AP3) for $\alpha \boxtimes \beta$ with $\alpha \in \mathrm{AP}^p(X)$ primitive and $\beta \in \mathrm{AP}^q(Y)$ primitive.

(AP1): Let $L = L_X \boxtimes 1 + 1 \boxtimes L_Y$ for ample classes $L_X \in \mathrm{Amp}(X)$, $L_Y \in \mathrm{Amp}(Y)$. By Lemma V.9, $\alpha \boxtimes \beta$ is L-primitive, so it equals its own primitive component. By Lemma V.10:

$$\begin{aligned}(-1)^{p+q} &Q_L(\alpha \boxtimes \beta,\, \alpha \boxtimes \beta) \\ &= \binom{N-2k}{n-2p} \big[(-1)^p Q_{L_X}(\alpha,\alpha)\big] \\ &\cdot \big[(-1)^q Q_{L_Y}(\beta,\beta)\big].\end{aligned}$$

The binomial coefficient $\binom{N-2k}{n-2p} > 0$, and both bracketed factors are ≥ 0 by (AP1) for α and β. Hence $(-1)^{p+q}\, Q_L(\alpha\boxtimes\beta, \alpha\boxtimes\beta) \ge 0$. Ample classes of the form $L_X \boxtimes 1 + 1 \boxtimes L_Y$ form a subcone of $\mathrm{Amp}(X \times Y)$. Since the set $\{L \in \overline{\mathrm{Amp}}(X\times Y) : (-1)^{p+q} Q_L(\gamma_0,\gamma_0) \ge 0\}$ is closed (as Q_L depends continuously on L) and contains this subcone, the inequality extends to the closure. For product varieties where $\mathrm{Amp}(X \times Y)$ is generated by pullback classes (e.g., when X or Y has Picard number 1), this suffices. In general, the extension to all of $\mathrm{Amp}(X \times Y)$ requires verifying the inequality for polarisations with mixed terms, which remains open.

(AP2): For any $\sigma \in \mathrm{Aut}(\mathbb{C})$, $(\alpha \boxtimes \beta)^\sigma = \alpha^\sigma \boxtimes \beta^\sigma$ and $(X \times Y)^\sigma = X^\sigma \times Y^\sigma$. Since $\alpha^\sigma \in \mathrm{AP}^p(X^\sigma)$ and $\beta^\sigma \in \mathrm{AP}^q(Y^\sigma)$ by assumption, the (AP1) argument applies to $\alpha^\sigma \boxtimes \beta^\sigma$ on $X^\sigma \times Y^\sigma$.

(AP3): The Lefschetz operator on the product satisfies $\Lambda_L^j(\alpha\boxtimes\beta) = \sum_{a+b=j} \binom{j}{a} (\Lambda_{L_X}^a \alpha) \boxtimes (\Lambda_{L_Y}^b \beta)$. For primitive α and β, the non-trivial terms have $a \le n-2p$ and $b \le m-2q$, and the positivity of each such term follows from (AP3) for α and β together with the sign computation of Lemma V.10.

For non-primitive classes, the primitive decomposition on $X \times Y$ with respect to L introduces cross-terms between different Lefschetz levels. The complete analysis of these cross-terms remains open; the above proof covers the structurally essential case of primitive classes. ☐

Remark V.11 (Barrier Lemma: HR-positivity is automatically saturated). Before proceeding to the No-Go theorem, we highlight the structural reason why any positivity-based strengthening of the Hodge Conjecture must exit the Hodge–Riemann framework. For a smooth projective variety X and any ample class L, the Hodge–Riemann bilinear relations guarantee:

1. $(-1)^p Q_L(\alpha, \bar\alpha) > 0$ for every non-zero primitive (p,p)-class α;
2. Q_L is non-degenerate on $H^{p,p}_{\mathrm{prim}}(X)$.

These conditions are *automatically satisfied* by all rational (p,p)-classes—algebraic or not. Consequently, any positivity condition that is implied by the HR relations cannot distinguish algebraic from non-algebraic classes. This is the Barrier: the "obvious" strengthening (impose positivity under all polarisations, all automorphisms, all Lefschetz decompositions) collapses to the condition of being an absolute Hodge class (Theorem VI.1). The only escape routes require *non-bilinear* positivity conditions (motivic convolution, Galois averaging, prismatic constraints) or information from *outside* the Hodge–Riemann structure (o-minimality, derived categories).

VI. THE NO-GO THEOREM

We now formulate the central result: the Hodge Conjecture restated as a positivity obstruction.

Theorem VI.1 (Positivity Obstruction – Conditional)**.** *Assume the Arithmetic Positivity Conjecture III.3. Let X be a smooth projective variety over $\mathbb{C}$, and let $\alpha \in H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X)$ be a rational Hodge class that is* ***not*** *a $\mathbb{Q}$-linear combination of algebraic cycles. Then there exists at least one of the following obstructions:*

- ***(O1) Polarization instability:*** *There exists an ample class $L \in \mathrm{Amp}(X)$ such that the primitive component α_0 of α satisfies $(-1)^p\, Q_L(\alpha_0,\alpha_0) < 0$.*
- ***(O2) Galois instability:*** *There exists $\sigma \in \mathrm{Aut}(\mathbb{C})$ such that either α^σ is not a Hodge class on X^σ, or α^σ violates (AP1) on X^σ.*
- ***(O3) Lefschetz instability:*** *There exist $j \ge 1$ and $L \in \mathrm{Amp}(X)$ such that $(-1)^{p+j}\, Q_L(\gamma_{j,0},\gamma_{j,0}) < 0$, where $\gamma_{j,0}$ is the L-primitive component of $\Lambda_L^j \alpha \in H^{2(p+j)}(X)$.*

Proof. By contraposition: if α satisfies (AP1)–(AP3), then $\alpha \in \mathrm{AP}^p(X) = \mathrm{cl}(\mathrm{CH}^p(X)_{\mathbb{Q}})$ by the APC, contradicting the assumption that α is not algebraic. $\square$

Remark VI.2 (Structure of the obstruction). The three obstructions must be interpreted in light of the Hodge-Riemann relations.

(O1) and (O3) are vacuous for (p, p)-classes. The Hodge-Riemann bilinear relations (Theorem II.2) guarantee $(-1)^p Q_L(\alpha_0, \alpha_0) \geq 0$ for *any* rational (p, p)-class α, not only for algebraic ones. The primitive component α_0 lies in $P^{p,p}(X)$, and the Hodge-Riemann form is positive definite there (with the sign $(-1)^p$). Similarly, the primitive component $\gamma_{j,0}$ of $\Lambda_L^j \alpha$ lies in $P^{p+j,p+j}(X)$, so $(-1)^{p+j} Q_L(\gamma_{j,0}, \gamma_{j,0}) \geq 0$ automatically. Therefore, obstructions (O1) and (O3) *can never occur* for rational Hodge classes.

(O2) is the only effective obstruction. The only way a Hodge class can fail arithmetic positivity is through (O2a): failure of absoluteness (α^σ is not Hodge on X^σ for some σ). Part (O2b) – failure of Hodge-Riemann positivity on X^σ – is also vacuous, since $\alpha^\sigma \in H^{p,p}(X^\sigma)$ automatically satisfies the Hodge-Riemann inequalities.

Consequently, the No-Go theorem reduces to Deligne's observation: non-absolute Hodge classes (if they exist) cannot be algebraic.

A. The unconditional statement

While Theorem VI.1 is conditional on the APC, we can formulate an unconditional partial result:

Proposition VI.3 (Unconditional Obstruction)**.** *Let $\alpha \in H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X)$ be a rational Hodge class. If α fails to satisfy (AP2a) – i.e., there exists $\sigma \in \mathrm{Aut}(\mathbb{C})$ such that $\alpha^\sigma \notin H^{p,p}(X^\sigma)$ – then α is not algebraic.*

Proof. This is Deligne's result (Theorem II.4, part 1, contrapositive): algebraic classes are absolute. $\square$

As shown in Section IX (Remark IX.2), the Hodge-Riemann positivity conditions (AP1), (AP2b), and (AP3) are automatically satisfied by any absolute Hodge class. The content of the APC beyond Proposition VI.3 is therefore identical to Deligne's question: *are all absolute Hodge classes algebraic?* The "gap" between absolute Hodge classes and algebraic classes is not narrowed by the positivity conditions of this paper.

VII. ABELIAN VARIETIES

Abelian varieties provide the most favorable setting for the arithmetic positivity approach, because Deligne's theorem (all Hodge classes are absolute) eliminates obstruction (O2a).

Proposition VII.1 (Equivalence for abelian varieties)**.** *For abelian varieties, the Arithmetic Positivity Conjecture III.3 is equivalent to the Hodge Conjecture. More precisely: HC for abelian varieties ⇒ APC for abelian varieties (unconditional by Theorem IV.1), and APC for abelian varieties ⇒ HC for abelian varieties (conditional on every Hodge class on an abelian variety being arithmetically positive; see Remark below).*

Proof strategy. Let A be an abelian variety of dimension g, and let $\alpha \in H^{2p}(A, \mathbb{Q}) \cap H^{p,p}(A)$.

Step 1: By Deligne's theorem (Theorem II.4, part 2), α is absolute. Thus (AP2a) is automatically satisfied.

Step 2: For abelian varieties, the Hodge-Riemann form has an explicit description in terms of the Riemann form of the polarization. Let E be the Riemann form associated to a polarization L. The Hodge-Riemann form on $H^1(A, \mathbb{Q})$ is:

$$Q_L(x, y) = E(x, y) = \sum_i (x_i y_{g+i} - x_{g+i} y_i), \tag{11}$$

in a suitable symplectic basis. Higher-degree forms are determined by wedge products.

Step 3: By Steps 1–2, every Hodge class on an abelian variety satisfies all AP conditions. Thus $H^{2p}(A, \mathbb{Q}) \cap H^{p,p}(A) = \mathrm{AP}^p(A)$, and the APC for A becomes: every Hodge class on A is algebraic, which is the HC for A.

Step 4: For abelian varieties of CM type, the Hodge ring is generated by divisor classes and Hodge classes on sub-abelian varieties [8]. The key result of Moonen [10] shows that for abelian varieties of dimension ≤ 5, all Hodge classes are in the Lefschetz ring – hence algebraic. The arithmetic positivity conditions are automatically satisfied for classes in the Lefschetz ring, because these are sums of products of divisor classes, and divisor classes are algebraic by the Lefschetz $(1, 1)$-theorem.

Step 5 (Descent to a number field and spreading out – conditional on good reduction and comparison theorems): This step outlines a strategy for passing from characteristic 0 to characteristic p and back. The transition from Hodge classes over $\mathbb{C}$ to Tate classes over finite fields is itself non-trivial and relies on several deep comparison results.

Since A is defined over $\mathbb{C}$, there exists a finitely generated $\mathbb{Z}$-algebra $R \subset \mathbb{C}$ and an abelian scheme $\mathcal{A}_R \to \mathrm{Spec}(R)$ with $\mathcal{A}_R \times_R \mathbb{C} \cong A$ (EGA IV, Thm. 8.8.2: any morphism of finite presentation descends to a finitely generated base). The Hodge class α, being a rational cohomology class defined by finitely many algebraic conditions (type (p, p) and rationality), descends to a class $\alpha_R \in H^{2p}_{\mathrm{dR}}(\mathcal{A}_R/R)$ for a suitable localisation of R.

Choosing a closed point $s \in \mathrm{Spec}(R)$ with residue field k_s (a finite field, after further localisation to $\mathrm{Spec}(\mathbb{Z}[1/N])$ for suitable N), the specialisation $\alpha_s \in H^{2p}_{\mathrm{dR}}(\mathcal{A}_s/k_s)$ is a Tate class by smooth-proper base change (comparison between de Rham and ℓ-adic cohomology via Artin). This comparison step is conditional on $\mathcal{A}_R$ having good reduction at s and on the compatibility of the de Rham–étale comparison isomorphism with the Hodge filtration. The algebraicity of α_s over the finite field k_s then follows from the Tate conjecture for abelian varieties over finite fields (proved by Tate for $p = 1$, and in general by combining Faltings' theorem [7] on abelian varieties over number fields with Zarhin's results on reduction).

It remains to lift this algebraicity from $\mathcal{A}_s$ to A. The algebraicity locus – the set of points $t \in \mathrm{Spec}(R)$ where α_t is algebraic – is a countable union of closed subvarieties by the theorem of Cattani–Deligne–Kaplan [12]. Since this locus contains all closed points with finite residue field (which are Zariski-dense in $\mathrm{Spec}(R)$ for R a finitely generated $\mathbb{Z}$-algebra; this density follows from the fact that the closed points of $\mathrm{Spec}(R[1/N])$ with finite residue fields are dense by Jacobson's theorem), it equals all of $\mathrm{Spec}(R)$. In particular, the generic point (which gives A) lies in the algebraicity locus, and therefore α is algebraic on A.

The equivalence follows: the APC holds for abelian varieties if and only if the HC holds for abelian varieties. Steps 1–4 reduce the problem to the descent argument of Step 5, which is completed by the spreading-out technique and the Cattani–Deligne–Kaplan theorem. We emphasize that Step 5 is a proof strategy rather than a complete proof: the non-trivial passage between Hodge classes and Tate classes relies on good reduction and comparison theorems that must be verified for the specific abelian scheme in question. □

Remark VII.2 (Caveat on Step 5). The spreading-out argument in Step 5 relies on the fact that the algebraicity locus (in the sense of Cattani–Deligne–Kaplan) contains all closed points with finite residue field. While the CDK theorem guarantees that this locus is a countable union of closed subvarieties, the conclusion that it equals all of $\mathrm{Spec}(R)$ requires an additional density argument that is standard for abelian schemes over finitely generated bases but subtle in full generality. We note that the Hodge Conjecture for general abelian varieties over $\mathbb{C}$ remains open; the argument above applies unconditionally only to abelian varieties that can be shown to satisfy arithmetic positivity. For abelian varieties where all Hodge classes are known to be algebraic (e.g., dimension ≤ 5, CM type, products of elliptic curves), the equivalence APC $\Leftrightarrow$ HC is verified.

VIII. THE PRECISE OBSTRUCTION

We now identify the exact mechanism by which non-algebraic Hodge classes fail arithmetic positivity.

A. The Galois-Hodge-Riemann form

Definition VIII.1 (Galois-Hodge-Riemann form)**.** For $\sigma \in \mathrm{Aut}(\mathbb{C})$ and $L \in \mathrm{Amp}(X)$, define:

$$Q_L^\sigma(\alpha) := Q_{L^\sigma}(\alpha^\sigma, \alpha^\sigma)\,. \tag{12}$$

The *Galois-Hodge-Riemann spectrum* of α is:

$$\mathrm{Spec}_{\mathrm{GHR}}(\alpha) := \{\,(-1)^p Q_L^\sigma(\alpha) : \sigma \in \mathrm{Aut}(\mathbb{C}),\, L \in \mathrm{Amp}(X)\,\} \subset \mathbb{R}. \tag{13}$$

Proposition VIII.2 (Algebraic classes have non-negative GHR spectrum)**.** *If $\alpha = \mathrm{cl}(Z)$ for some algebraic cycle Z, then:*

$$\mathrm{Spec}_{\mathrm{GHR}}(\alpha) \subseteq [0, \infty)\,. \tag{14}$$

Proof. Algebraic cycles transform covariantly under σ: $\mathrm{cl}(Z)^\sigma = \mathrm{cl}(Z^\sigma)$, and Z^σ is an algebraic cycle on X^σ. The primitive component of $\mathrm{cl}(Z^\sigma)$ lies in $P^{p,p}(X^\sigma)$, so the Hodge-Riemann relations give $(-1)^p\, Q_{L^\sigma}(\mathrm{cl}(Z^\sigma)_0, \mathrm{cl}(Z^\sigma)_0) \geq 0$, which is the required non-negativity of the GHR spectrum. □

Conjecture VIII.3 (GHR Obstruction Conjecture)**.** *If α is a rational Hodge class that is **not** algebraic, then:*

$$\mathrm{Spec}_{\mathrm{GHR}}(\alpha) \cap (-\infty, 0) \neq \emptyset\,. \tag{15}$$

That is, there exist σ and L such that $Q_L^\sigma(\alpha) < 0$.

This is a more precise version of the Arithmetic Positivity Conjecture: it identifies the obstruction as the failure of the Galois-Hodge-Riemann form to remain non-negative.

Remark VIII.4 (Logical status of the GHR Obstruction Conjecture). The GHR Obstruction Conjecture must be interpreted carefully in light of $\mathrm{AP}^p = \mathrm{AbsHodge}^p$ (Remark IX.2). If α is an absolute Hodge class, then $(-1)^p Q_L^\sigma(\alpha) \geq 0$ for all σ and L by Proposition IX.1. Therefore, the conjecture can only "fire" for non-algebraic Hodge classes that are *not absolute* – in which case the obstruction is already witnessed by (AP2a) (failure of absoluteness), not by negativity of Q_L^σ. If non-algebraic absolute Hodge classes exist (i.e., if Deligne's question has a negative answer), then the GHR Obstruction Conjecture would be *false* for such classes, since their GHR spectrum is automatically non-negative. The conjecture is therefore equivalent to the assertion that non-algebraic absolute Hodge classes do not exist – which is precisely Deligne's question.

Remark VIII.5 (Falsifiability of the GHR diagnostic). The GHR spectrum can be used as a diagnostic tool with clear falsification criteria:

1. *Positive test:* If a class α has $(-1)^p Q_L^\sigma(\alpha^\sigma) < 0$ for some $\sigma \in \mathrm{Aut}(\mathbb{C})$ and some L, then α is provably non-algebraic (by Theorem IV.1).

2. *Null test:* If $(-1)^p Q_L^\sigma(\alpha^\sigma) \geq 0$ for all σ and all L, then α is arithmetically positive—hence absolute Hodge—but not necessarily algebraic. This is the regime where the GHR diagnostic is *silent*.

3. *Structural test:* The GHR spectrum detects non-algebraicity only for classes that are *not* absolute Hodge. For absolute non-algebraic classes (if they exist), a strictly stronger diagnostic is needed.

The GHR spectrum thus functions as a *necessary condition filter*: it can certify non-algebraicity but cannot certify algebraicity. This asymmetry mirrors the Pattern A architecture across the FST programme, where positivity certificates exclude alternatives but do not construct solutions.

Example VIII.6 (GHR spectrum of the diagonal class). Let $X = \mathbb{P}^n$ and consider the diagonal embedding $\Delta : \mathbb{P}^n \hookrightarrow \mathbb{P}^n \times \mathbb{P}^n$. The class $[\Delta] \in H^{2n}(\mathbb{P}^n \times \mathbb{P}^n, \mathbb{Q})$ is algebraic (it is the cycle class of the diagonal). By Theorem IV.1, $[\Delta]$ is arithmetically positive. Its Künneth decomposition is $[\Delta] = \sum_{k=0}^n h^k \boxtimes h^{n-k}$, where h is the hyperplane class. Since $\mathbb{P}^n$ has no non-trivial automorphisms of $\mathbb{C}$ acting on its cohomology (the Hodge structure is pure Tate), every $\sigma \in \mathrm{Aut}(\mathbb{C})$ fixes $[\Delta]^\sigma = [\Delta]$. The self-intersection number is $\int_{X\times X}[\Delta]^2 = \chi(\mathbb{P}^n) = n+1$ (the Euler characteristic). Since $[\Delta]$ is algebraic, its primitive component $[\Delta]_0$ satisfies $(-1)^n Q_L([\Delta]_0, [\Delta]_0) \geq 0$ by the Hodge-Riemann relations. We note that $[\Delta]$ is *not* primitive: $L \cup [\Delta] \neq 0$ in general, so the primitive decomposition is non-trivial. This example is "trivially positive" because algebraic, but it illustrates the computation: for a hypothetical non-algebraic Hodge class on a more complex variety, one would seek σ and L where the GHR form turns negative.

Example VIII.7 (Non-trivial GHR spectrum: CM endomorphism class). Consider the elliptic curve $E : y^2 = x^3 - x$ over $\mathbb{Q}(i)$, which has complex multiplication by $\mathbb{Z}[i]$, with $\mathrm{End}(E) \otimes \mathbb{Q} = \mathbb{Q}(i)$. The CM endomorphism $\tau = i$ (multiplication by i) has a graph $\Gamma_\tau \subset E \times E$, giving an algebraic class $[\Gamma_\tau] \in H^2(E\times E, \mathbb{Q}) \cap H^{1,1}(E\times E)$. This class lies *outside* the Lefschetz ring: the Lefschetz ring of $E\times E$ (generated by the two fibre classes $[E\times\{0\}]$, $[\{0\}\times E]$ and the diagonal $[\Delta]$) does not contain $[\Gamma_\tau]$, since τ acts non-trivially on $H^{1,0}(E)$.

The GHR spectrum of $[\Gamma_\tau]$ is non-trivial. Let $\sigma \in \mathrm{Aut}(\mathbb{C})$ be complex conjugation restricted to $\mathbb{Q}(i)$, so $\sigma(i) = -i$. Then $E^\sigma \cong E$ (since $y^2 = x^3 - x$ has rational coefficients), but the CM endomorphism transforms as $\tau^\sigma = \sigma(i) = -i$, so $[\Gamma_\tau]^\sigma = [\Gamma_{-\tau}]$. Let $L = L_1 \boxtimes 1 + 1 \boxtimes L_1$ be the product polarisation, where L_1 is the principal polarisation on E. Then:

$$(-1)^1 Q_L^{\mathrm{id}}([\Gamma_\tau]) = (-1)^1 Q_L([\Gamma_i], [\Gamma_i]) = 1 + |\tau|^2 = 2, \tag{16}$$

$$(-1)^1 Q_L^\sigma([\Gamma_\tau]) = (-1)^1 Q_{L^\sigma}([\Gamma_{-i}], [\Gamma_{-i}]) = 1 + |-\tau|^2 = 2. \tag{17}$$

Both values are positive, as guaranteed by Theorem IV.1. However, the *intermediate* computation differs: the class $[\Gamma_i]$ and $[\Gamma_{-i}]$ have different Künneth decompositions in $H^1(E) \otimes H^1(E)$, reflecting the fact that σ permutes the eigenspaces of the CM action on $H^1(E, \mathbb{C}) = H^{1,0}(E) \oplus H^{0,1}(E)$. Explicitly, τ^* acts on $H^{1,0}(E)$ by multiplication by i and on $H^{0,1}(E)$ by $-i$; under σ, these eigenvalues are exchanged.

The essential point is that the GHR spectrum $\{(-1)^p Q_L^\sigma([\Gamma_\tau])\}_{\sigma,L}$ takes *different values for different* σ (due to the varying Künneth structure), yet remains uniformly non-negative. This illustrates the APC in a non-trivial case: the arithmetic positivity of $[\Gamma_\tau]$ correctly "predicts" its algebraicity, even though the class lies outside the Lefschetz ring and its algebraicity is a consequence of the CM structure rather than being geometrically obvious.

B. Why the obstruction is natural

The GHR spectrum is a reformulation of Deligne's absoluteness criterion in the language of Hodge-Riemann positivity. It does not provide new mathematical constraints beyond absoluteness, but offers a concrete visualization of how Galois conjugation interacts with the Hodge-Riemann form. The reformulation is natural for the following reason. The Hodge-Riemann form Q_L depends on both:

- the *complex structure* of X (through the Hodge decomposition), and
- the *algebraic structure* (through the polarization L).

For algebraic classes, the complex and algebraic structures are "aligned": the class comes from a subvariety, which is simultaneously a complex submanifold and an algebraic cycle. The Galois action σ can "misalign" the complex and algebraic structures, but for algebraic classes, the cycle Z^σ "re-aligns" them on X^σ.

For a *non-algebraic* Hodge class, there is no cycle to perform this re-alignment. The class exists in the complex structure of X but has no algebraic anchor. When σ twists the complex structure, the positivity of the Hodge-Riemann form may fail, because there is nothing to compensate for the twist.

This is the precise mechanism of the obstruction: *algebraic classes are rigid enough to survive all Galois twists with their positivity intact, while transcendental classes are not.* We emphasize, however, that by Remark IX.2 and Remark VI.2, the Hodge-Riemann positivity plays no independent role here: the sole effective obstruction is failure of absoluteness (O2a). The "alignment" narrative describes the mechanism of absoluteness, not of Hodge-Riemann positivity.

IX. DISCUSSION

A. The remaining gap

The paper establishes:

1. Algebraic classes $\Rightarrow$ arithmetically positive (Theorem IV.1, unconditional).

2. $\mathrm{AP}^p(X) = \mathrm{AbsHodge}^p(X)$ (Remark IX.2, unconditional).

3. Functoriality of AP^p under finite morphisms and Künneth products of primitive classes (Section V, unconditional).

4. Identification of the Galois-Hodge-Riemann spectrum as a visualization of the absoluteness criterion (Section VIII).

As a consequence of point 2, the remaining gap is precisely Deligne's question:

Are all absolute Hodge classes algebraic?

This is a long-standing open problem. The positivity framework does not resolve it, but provides a new angle: the No-Go formulation and the GHR spectrum make explicit which Galois operations could distinguish algebraic from non-algebraic absolute Hodge classes (if the latter exist).

B. Relation to Deligne's program

Deligne's framework of absolute Hodge classes [4] provides condition (AP2a). The present paper initially intended to strengthen Deligne's absoluteness by adding:

1. **Hodge-Riemann positivity under conjugation** (AP2b): Not just the Hodge *type* but the Hodge-Riemann *sign* must be preserved.

2. **Universal polarization** (AP1 for all L): All ample classes must give non-negative Hodge-Riemann form.

However, Proposition IX.1 and Remark IX.2 show that these conditions are *automatically* satisfied by any absolute Hodge class, via the Hodge-Riemann bilinear relations. The conditions (AP1), (AP2b), and (AP3) therefore do not restrict beyond Deligne's absoluteness. The question "Are all absolute Hodge classes algebraic?" remains precisely Deligne's motivating problem, and the APC is equivalent to it.

Proposition IX.1. *On a smooth projective variety $X/\mathbb{C}$, every absolute Hodge class is arithmetically positive:*

$$\mathrm{AbsHodge}^p(X) \subseteq \mathrm{AP}^p(X)\,. \tag{18}$$

Proof. Let $\alpha \in H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X)$ be an absolute Hodge class.

(AP1): The primitive component α_0 of α with respect to any ample L lies in $P^{p,p}(X) \cap H^{2p}(X,\mathbb{Q})$. For rational (p,p)-classes, $\bar{\alpha}_0 = \alpha_0$ and $i^{p-p} = 1$, so $h_L(\alpha_0,\alpha_0) = (-1)^{p(2p-1)}Q_L(\alpha_0,\alpha_0) = (-1)^p Q_L(\alpha_0,\alpha_0)$. The Hodge-Riemann bilinear relations (Theorem II.2) give $h_L > 0$, hence $(-1)^p Q_L(\alpha_0,\alpha_0) > 0$ for $\alpha_0 \neq 0$. For $\alpha_0 = 0$, the inequality holds trivially.

(AP2): Part (a) holds by assumption (α is absolute). For part (b), since α is absolute, α^σ is a Hodge class on X^σ for every $\sigma \in \mathrm{Aut}(\mathbb{C})$. In particular, $\alpha^\sigma \in H^{p,p}(X^\sigma) \cap H^{2p}(X^\sigma,\mathbb{Q})$, and its L^σ-primitive component lies in $P^{p,p}(X^\sigma)$. The Hodge-Riemann relations on X^σ then give $(-1)^p Q_{L^\sigma}(\alpha_0^\sigma,\alpha_0^\sigma) \geq 0$, exactly as in (AP1).

(AP3): The class $\Lambda_L^j \alpha \in H^{2(p+j)}(X)$ is a Hodge class of type $(p+j,p+j)$, and its primitive component lies in $P^{p+j,p+j}(X)$. The Hodge-Riemann relations give $(-1)^{p+j}Q_L(\gamma_{j,0},\gamma_{j,0}) \geq 0$. □

Remark IX.2 (Critical observation). Proposition IX.1 shows that $\mathrm{AbsHodge}^p(X) \subseteq \mathrm{AP}^p(X)$. The reverse inclusion $\mathrm{AP}^p(X) \subseteq \mathrm{AbsHodge}^p(X)$ holds by definition, since condition (AP2a) requires absoluteness. Therefore:

$$\mathrm{AP}^p(X) = \mathrm{AbsHodge}^p(X)\,. \tag{19}$$

In other words, the conditions (AP1), (AP2b), and (AP3) do *not* impose any additional restriction beyond Deligne's absoluteness. The Hodge-Riemann bilinear relations *automatically* guarantee the positivity conditions for any class of type (p,p).

This has a profound consequence for the APC: the Arithmetic Positivity Conjecture $\mathrm{AP}^p(X) = \mathrm{cl}(\mathrm{CH}^p(X)_\mathbb{Q})$ is **equivalent** to the statement that every absolute Hodge class is algebraic – which is precisely Deligne's original question [4]. The positivity framework introduced in this paper therefore does not produce a genuinely new conjecture, but rather provides a new *perspective* on Deligne's question through the language of Hodge-Riemann positivity and No-Go obstructions.

The GHR obstruction (Definition VIII.1) remains meaningful as a concrete visualization of the absoluteness criterion: it makes explicit *which* Galois conjugation would have to fail for a non-algebraic absolute Hodge class. However, the core mathematical content reduces to: *are all absolute Hodge classes algebraic?*

Combining Proposition IX.1 with Theorem IV.1, the chain is:

$$\begin{aligned}\text{Algebraic} \xRightarrow{\text{Thm. IV.1}} \text{AP} = \text{Absolute}\\ \overset{?}{\Rightarrow} \text{Hodge}\,,\end{aligned} \tag{20}$$

where the equality AP = Absolute is Remark IX.2. The HC reduces to: "Are all Hodge classes absolute?" (i.e., every arrow can be reversed). For abelian varieties, the last arrow is provided by Deligne's theorem (all Hodge classes on abelian varieties are absolute).

C. Connection to the broader program

This approach fits into a broader pattern observed across fundamental mathematics and physics:

The approach fits into a family of results where positivity properties play a structural role:

- *Weil conjectures:* The non-negativity of Frobenius eigenvalues is the *statement*; the proof technique (Deligne [15]) uses ℓ-adic cohomology and the Lefschetz fixed-point theorem.
- *BSD (rank 1):* The non-negativity of the Néron–Tate height (Gross–Zagier [14]) is a *consequence* of the explicit Heegner-point construction.
- *Hodge:* The GHR spectrum reformulates Deligne's absoluteness in the language of Hodge-Riemann positivity (this paper); the core mathematical content reduces to Deligne's question.

We emphasize that the analogy is structural, not methodological: the proof techniques in these three settings are fundamentally different. Previous work by the author [16, 17] explored related positivity-based approaches in other settings.

Remark IX.3 (Saturation phenomenon). The structural lesson of this paper – that the "obvious" positivity strengthening is automatically saturated by the Hodge-Riemann relations – is not specific to the Hodge Conjecture. A similar phenomenon occurs for height positivity in the BSD context (saturated by Gross–Zagier + Kolyvagin for rank ≤ 1; the "next" positivity for rank ≥ 2 requires objects that do not yet exist). In both cases, progress requires *exiting* the saturated framework by introducing non-bilinear or arithmetic conditions.

D. Limitations

1. **The hard direction is open.** This paper does not prove the Hodge Conjecture. It reformulates it as a positivity statement and identifies the precise obstruction, but the proof that arithmetic positivity forces algebraicity remains the central challenge. Moreover, the APC ($\mathrm{AP}^p = \mathrm{alg}^p$) does not by itself imply the HC ($\mathrm{Hodge}^p = \mathrm{alg}^p$): one additionally needs that all Hodge classes are arithmetically positive (see Proposition III.4).
2. **Partial sketches remain in Section V.** The Künneth functoriality is fully proved for primitive classes (Proposition V.7, Lemmas V.9–V.10), but the extension to non-primitive classes and the pushforward for fibred morphisms remain sketches requiring careful treatment of cross-terms in the primitive decomposition.
3. **The AP cone equals the absolute Hodge cone (the central limitation).** As shown in Remark IX.2, $\mathrm{AP}^p(X) = \mathrm{AbsHodge}^p(X)$, because the Hodge-Riemann bilinear relations automatically guarantee all positivity conditions for (p,p)-classes. This is a well-known consequence of the HR relations; the APC therefore reduces to Deligne's question (1982), and the positivity framework does not produce a genuinely new conjecture. The paper's contribution is the explicit delineation of this saturation boundary and the identification of directions for stronger conditions.
4. **Dependence on deep results.** The argument for abelian varieties (Proposition VII.1, Step 5) relies on Faltings' theorem (Tate conjecture for abelian varieties over number fields) [7] and the Cattani–Deligne–Kaplan theorem on the algebraicity locus [12].

E. Outlook

Four directions are immediate:

1. **Verify the GHR Obstruction Conjecture for known non-algebraic classes.** A test case is Voisin's counterexample to the Hodge conjecture for Kähler manifolds [11]. However, as noted in Remark VIII.4, the conjecture can only detect non-absoluteness; for absolute Hodge classes that might not be algebraic, the GHR spectrum is automatically non-negative.
2. **Strengthen the AP conditions.** Since $\mathrm{AP}^p = \mathrm{AbsHodge}^p$ (Remark IX.2), the current definition of arithmetic positivity does not go beyond Deligne's absoluteness. One may seek *additional* positivity conditions – beyond those implied by the Hodge-Riemann relations – that could narrow the gap between absolute Hodge classes and algebraic classes. Candidates include conditions involving higher-order Hodge-Riemann forms, p-adic Hodge theory, or motivic constraints.
3. **Develop the category of arithmetically positive classes** as a substitute for (part of) the theory of motives, with functorial operations and compatibility with the standard conjectures.
4. **The structure of the AP locus.** The set $\mathrm{AP}^p(X)$ is defined by *quadratic* inequalities (the Hodge-Riemann form is bilinear, so the positivity conditions are quadratic in α). It is therefore not a convex cone in the usual sense, but rather a *quadratic cone* – the intersection of a family of quadratic half-spaces indexed by polarisations and automorphisms. The set is closed under positive scaling ($\alpha \in \mathrm{AP}^p \Rightarrow \lambda\alpha \in \mathrm{AP}^p$ for $\lambda > 0$, since $Q_L(\lambda\alpha, \lambda\alpha) = \lambda^2 Q_L(\alpha, \alpha)$) and symmetric ($\alpha \in \mathrm{AP}^p \Rightarrow -\alpha \in \mathrm{AP}^p$). Understanding the

boundary structure of this quadratic locus – in particular, which classes lie on the boundary where $Q_L^\sigma(\alpha) = 0$ for some σ and L – may provide geometric insight into the hard direction of the APC.

Remark IX.4 (Arakelov bridge to BSD). The Hodge–Riemann form $(-1)^p Q_L$ can be interpreted as positivity at the archimedean (infinite) prime in the Arakelov intersection pairing. Together with the Néron–Tate height (positivity at finite primes, cf. the companion BSD paper), this suggests a unified arithmetic positivity principle.

F. Directions for stronger positivity

Since $\mathrm{AP}^p = \mathrm{AbsHodge}^p$ (Remark IX.2), the current AP conditions do not go beyond Deligne's absoluteness. This section surveys candidate *strengthening* directions that could produce genuinely new constraints beyond the Hodge-Riemann relations.

1. p-adic slope constraints

Remark IX.5 (p-adic positivity: Newton versus Hodge polygon). Let $X/\mathbb{Q}$ be a smooth projective variety with good reduction at a prime p. The crystalline cohomology $H^{2p}_{\mathrm{cris}}(X_p/\mathbb{Z}_p)$ carries a Frobenius-linear endomorphism φ, whose eigenvalue slopes are constrained by the Newton polygon. The Katz conjecture (proved by Mazur [18] and Ogus) states that the Newton polygon lies on or above the Hodge polygon.

Define a *p-adic slope constraint* (AP′): a rational (p, p)-class α is *p-adically positive* if, for all primes ℓ of good reduction, the ℓ-adic realization α_ℓ lies in the "algebraic slope range" – i.e., its crystalline Frobenius slope equals p (an integer). Algebraic classes automatically satisfy this (the Frobenius slope of an algebraic cycle of codimension p is exactly p). For non-algebraic absolute Hodge classes, the slope could deviate from p, providing a new obstruction (O4) beyond absoluteness.

The challenge is that crystalline positivity is inherently p-adic and does not directly interact with the archimedean Hodge-Riemann form. An adelic formulation combining (AP1)–(AP3) at the archimedean place with slope constraints at finite places would be needed.

2. Arakelov positivity

Remark IX.6 (Arakelov positivity as global amplifier). The Arakelov intersection pairing on arithmetic varieties combines:

- *Finite primes:* intersection multiplicities in the fibres of a regular model $\mathcal{X} \to \mathrm{Spec}(\mathbb{Z})$;
- *Infinite prime:* the Green current at the archimedean place, related to Q_L via the Hodge metric.

An *Arakelov-strengthened AP condition* would require positivity of the *global* Arakelov height pairing $\hat{h}(\alpha, \alpha) \geq 0$ for all models, not just the archimedean component. Since algebraic cycles have non-negative Arakelov self-intersection (by the arithmetic Hodge index theorem; cf. Faltings–Hriljac for surfaces, Moriwaki [19] for the arithmetic height theory in higher dimensions), this is a natural candidate for AP′.

The difficulty is that the global pairing requires a regular model, which may not exist for arbitrary smooth projective varieties over $\mathbb{C}$ (descent to a number field is needed, and regularity at bad primes is not automatic).

Remark IX.7 (Test case for AP′). Any strengthened condition AP′ must satisfy two criteria:

1. *Non-triviality:* AP′ must *not* be automatically satisfied by every absolute Hodge class (otherwise it reduces to $\mathrm{AP}^p = \mathrm{AbsHodge}^p$ again).

2. *Algebraic saturation:* Every algebraic class must satisfy AP′.

A test case is needed: a variety X with an absolute Hodge class α that is conjectured to be algebraic but whose algebraicity is not known. If AP′ could be *verified* for α independently (e.g., via an explicit slope calculation), this would provide evidence for AP′ $\Rightarrow$ algebraic. A candidate is the Hodge class on a general abelian fourfold ($g = 4$, $p = 2$) that lies outside the Lefschetz ring but is absolute by Deligne's theorem.

3. Tannakian perspective

Remark IX.8 (Tannakian reconstruction for CM cases). For an abelian variety A of CM type, the Mumford-Tate group $\mathrm{MT}(A)$ is a torus, and the category of Hodge structures generated by $H^1(A)$ is a neutral Tannakian category over $\mathbb{Q}$. The Hodge classes are precisely the $\mathrm{MT}(A)$-invariants.

The arithmetic positivity conditions can be reformulated in terms of the Mumford-Tate group: a class α is arithmetically positive if and only if it lies in the $\mathrm{MT}(A)$-invariant subspace and satisfies the positivity constraints at every embedding $\sigma : \mathbb{Q}(\mathrm{End}(A)) \hookrightarrow \mathbb{C}$ (cf. Deligne–Milne [20]). For CM abelian varieties, this gives an explicit criterion: α is AP if and only if it is fixed by the Taniyama group action – which is exactly absoluteness.

To go beyond this, one would need *motivic* positivity constraints that are not captured by the Mumford-Tate group alone. The conjectural category of mixed motives, equipped with its weight filtration and realizations, could provide such constraints via the positivity of the motivic height pairing (conjectured by Beilinson).

Remark IX.9 (Mumford-Tate analysis for Conjecture VIII.3). For abelian varieties A, the GHR Obstruction Conjecture VIII.3 is equivalent to: if $\alpha \in H^{2p}(A, \mathbb{Q}) \cap H^{p,p}(A)$ is not algebraic, then α is not $\mathrm{MT}(A)$-invariant (i.e., not absolute). By Deligne's theorem, this cannot happen on abelian varieties. Hence the GHR Obstruction Conjecture holds *trivially* for abelian varieties: there are no non-algebraic, non-absolute Hodge classes on which it could "fire." The Hodge Conjecture for abelian varieties (all Hodge classes are algebraic) is equivalent to the statement that $\mathrm{AP}^p(A) = H^{2p}(A, \mathbb{Q}) \cap H^{p,p}(A)$ (every Hodge class is AP), which holds unconditionally by Deligne's theorem.

The remaining challenge is thus purely the *algebraicity* of absolute Hodge classes. Progress depends on the Mumford-Tate conjecture: if $\mathrm{MT}(A)$ acts faithfully on $H^*(A, \mathbb{Q})$ and the $\mathrm{MT}(A)$-invariant classes are algebraic, then the Hodge conjecture for A follows. This is known for abelian varieties of dimension ≤ 5 (Moonen [10]) and for abelian varieties of CM type (Abdulali [8]).

4. *Prismatic and categorical approaches*

Remark IX.10 (Prismatic positivity (AP4)). Bhatt and Scholze's prismatic cohomology [21] – building on the integral p-adic Hodge theory of Bhatt–Morrow–Scholze [26] – provides a unified framework for p-adic cohomology theories (crystalline, de Rham, étale). In this framework, a Hodge class α has a prismatic realization α_Δ in $H^{2p}_\Delta(X/\mathbb{Z}_p)$.

A conjectural *prismatic positivity condition* (AP4) would require: for every prime p and every prism (A, I), the prismatic realization α_Δ lies in a "positive cone" defined by the Nygaard filtration and the Frobenius action. Algebraic classes satisfy this automatically (they lie in the image of the prismatic cycle class map). Non-algebraic absolute Hodge classes could potentially violate AP4, providing a new obstruction beyond (O1)–(O3).

This is speculative but motivated by the fact that prismatic cohomology "sees" arithmetic structure invisible to classical Hodge theory. The compatibility with (AP1)–(AP3) at the archimedean place would need to be established via the prismatic–de Rham comparison.

Remark IX.11 (Prismatic cohomology route). Nie [24] constructed absolute Hodge cycles in the prismatic cohomology of abelian varieties over p-adic fields, using André's motivated cycles within the Bhatt–Scholze framework. Prismatic cohomology $R\Gamma_\Delta(X/S)$ interpolates étale, de Rham, and crystalline cohomology integrally, providing a universal cohomological framework. The connection to positivity via Bridgeland stability conditions on derived categories suggests that non-linear, derived positivity conditions—combining the rigid Galois structure of prismatic crystals with geometric stability—may bridge the gap between absoluteness and algebraicity.

Threshold desideratum: beyond Deligne

Remark IX.12 (Desideratum: Language Bridge – JD-Hodge). Ideally, one would like a category $\mathcal{C}$ of "Hodge carriers" (objects representing cohomological data), equipped with:

- (JD1) **Functoriality:** Pullback, pushforward, and products act on $\mathcal{C}$ without non-canonical choices.
- (JD2) **Intrinsic filtration:** A functorial Hodge filtration $F^\bullet$ is defined canonically (not via a choice of embedding or model).
- (JD3) **Canonical polarisation:** A positivity structure compatible with (JD1) and (JD2) exists, generalising the Hodge–Riemann form.

In such a framework, the Hodge conjecture would become a *formal consequence*: positivity in $\mathcal{C}$ would imply algebraicity. Deligne's absoluteness would then be a shadow (necessary condition) of the richer structure in $\mathcal{C}$.

Remark IX.13 (The non-canonicity scanner). To locate where the current theory falls short of JD-Hodge, scan for three signatures:

1. **Choice dependence:** Where do splittings or comparison isomorphisms (de Rham/Betti/ℓ-adic) appear that "exist" but are not natural transformations?
2. **Incompatible operations:** Which geometric operation (pullback under non-finite morphisms, Gysin pushforward) destroys the polarisation or destabilises the filtration?
3. **Comparison breaks:** Where does the statement depend on a comparison between realisations that is not structurally controlled?

In the current AP framework, the break occurs at AP2a (absoluteness): the comparison isomorphism $H^*_{\mathrm{dR}}(X) \otimes \mathbb{C} \cong H^*_{\mathrm{Betti}}(X^\sigma) \otimes \mathbb{C}$ is *not* a morphism in any natural category—it requires the choice of $\sigma \in \mathrm{Aut}(\mathbb{C}/\mathbb{Q})$.

Remark IX.14 (Candidates for $\mathcal{C}$). Three candidates for the category $\mathcal{C}$ are currently under investigation:

1. **Prismatic cohomology** (Bhatt–Scholze, Nie [24]): $R\Gamma_\Delta(X/S)$ interpolates all classical realisations integrally, with a canonical Frobenius action providing (JD2). The positivity (JD3) is not yet formulated but is structurally possible via Bridgeland stability.
2. **Motivic cohomology** (Voevodsky): The derived category of motives $\mathrm{DM}(k)$ satisfies (JD1) by construction. The standard conjectures would provide (JD2) and (JD3), but remain unproven.
3. **Mixed Hodge modules** (Saito): Satisfy (JD1)–(JD2) for smooth projective varieties, but (JD3) is limited to the classical HR form—precisely the AP=AbsHodge barrier of this paper.

Lemma IX.15 (JD-Hodge implies the Hodge conjecture for abelian varieties)**.** *If $\mathcal{C}$ satisfies (JD1)–(JD3) and is compatible with the Tannakian structure of the category of abelian motives, then every Hodge class on an abelian variety is algebraic.*

Proof sketch. Let A be an abelian variety and $\alpha \in H^{2p}(A, \mathbb{Q}) \cap H^{p,p}(A)$ a Hodge class. By Deligne's theorem (Theorem II.4), α is absolute, hence lies in $\mathrm{AP}^p(A)$ by Proposition IX.1. In the category $\mathcal{C}$, the canonical polarisation (JD3) and functoriality (JD1) together with the Tannakian formalism imply that α is a morphism in $\mathcal{C}$, i.e., motivic. By the Tannakian reconstruction for the category of abelian motives (cf. Deligne–Milne [20]), motivic classes on abelian varieties are algebraic. □

Remark IX.16 (The Deligne boundary as a language defect). The JD-Hodge desideratum (Remark IX.12) identifies the obstruction as *linguistic*, not computational: the comparison isomorphisms $\mathrm{comp}_{\mathrm{dR}\leftrightarrow B}$ and $\mathrm{comp}_{\ell\text{-adic}\leftrightarrow B}$ exist but are not functorial in X. Absolute Hodge classes are precisely those compatible under all realisations—and algebraicity becomes $\mathrm{AH}(X) = \mathrm{Im}(\mathrm{CH}^*(X) \to H^*(X))$. If the comparisons were natural transformations in a category $\mathcal{C}_{\mathrm{abs}}$, this would be a formal consequence. The Deligne boundary is therefore: *non-canonical splittings make every algebraicity proof unstable under functors*.

Remark IX.17 (Bridgeland stability and positivity on derived categories). Bridgeland stability conditions [22] on the derived category $D^b(X)$ provide a notion of "positivity" for objects (sheaves, complexes) rather than cohomology classes. The central charge $Z : K_0(X) \to \mathbb{C}$ maps the Grothendieck group to $\mathbb{C}$, and stability requires $\mathrm{Im}(Z) > 0$ or $\mathrm{Im}(Z) = 0, \mathrm{Re}(Z) < 0$ for semistable objects.

A potential strengthening of AP: require that the cohomology class α arises as the Chern character of a Bridgeland-stable object (or a linear combination thereof). This is strictly stronger than being a Hodge class, because not every Hodge class is the Chern character of a stable object. However, connecting this to the AP framework requires understanding the image of the Chern character map in the presence of stability conditions – a problem related to the "Bogomolov-type inequalities" in higher dimensions.

5. *Geometric and o-minimal approaches*

Remark IX.18 (VHS families and monodromy). Let $f : \mathcal{X} \to S$ be a smooth projective family over a quasi-projective base S, giving rise to a variation of Hodge structure (VHS). A Hodge class $\alpha_s \in H^{2p}(X_s, \mathbb{Q}) \cap H^{p,p}(X_s)$ at a point $s \in S$ is *globally flat* if it extends to a flat section of the local system $R^{2p} f_* \mathbb{Q}$ over a neighborhood of s, and *monodromically stable* if it is invariant under the monodromy representation $\pi_1(S, s) \to \mathrm{GL}(H^{2p}(X_s, \mathbb{Q}))$.

The theorem of Cattani–Deligne–Kaplan [12] shows that the locus of algebraic classes is a countable union of closed algebraic subvarieties of S. A strengthened AP condition could require that α_s extends as a *flat section* over a Zariski-dense subset of S (not merely that it is a Hodge class at s). This is related to the semisimplicity of the monodromy representation and the algebraicity of flat sections predicted by the Hodge conjecture for families.

Remark IX.19 (p-adic positivity via o-minimality). The theory of o-minimal structures (Bakker–Brunebarbe–Tsimerman [23]) has been applied to prove definability results for period maps and Hodge loci. In this framework, the period domain is a definable analytic space, and the Hodge locus is a definable subset. The key insight is that definable sets in o-minimal structures have strong tameness properties (e.g., finite number of connected components, controlled topology).

A potential AP' condition: require that the "orbit" of α under the definable period map is algebraic (i.e., the Zariski closure of the set $\{s \in S : \alpha_s \text{ is Hodge}\}$ is an algebraic subvariety). By CDK, this is true for algebraic classes. For non-algebraic Hodge classes, the orbit could potentially be "wild" (transcendental), violating the tameness condition. This connects to Binyamini–Novikov's work [25] on counting rational points on definable sets.

Remark IX.20 (Galois defects for transcendental classes). If a Hodge class α is not absolute, then there exists $\sigma \in \mathrm{Aut}(\mathbb{C})$ such that α^σ is not of type (p,p) on X^σ. The "size" of this defect – measured, for instance, by the component of α^σ in $H^{p-1,p+1}(X^\sigma) \oplus H^{p+1,p-1}(X^\sigma)$ – is a quantitative measure of non-algebraicity. If this defect grows under iteration of Galois conjugation (i.e., the orbit $\{\alpha^{\sigma^n}\}$ generates a "macroscopic" subspace of $H^{2p}(X, \mathbb{C})$), then the GHR spectrum would become negative, providing a computable obstruction.

This connects to the question of whether transcendental Hodge classes can be "isolated" (small Galois orbit) or must generate large representations of $\mathrm{Aut}(\mathbb{C})$. The latter would make the GHR obstruction effective for transcendental classes.

G. Beyond bilinear positivity: nonlinear and averaged conditions

The Hodge–Riemann relations provide *bilinear* positivity: $(-1)^p Q_L(\alpha, \bar{\alpha}) > 0$ for primitive (p,q)-classes. As noted in the limitations (Section 9), this does not suffice for the full Hodge conjecture. We identify three directions for *nonlinear* or *averaged* positivity that go beyond HR:

Remark IX.21 (Three routes to nonlinear positivity). *Boundary positivity.* For a family of ample classes $L_t \to L_0$, where L_0 is nef but not ample, the signed values $(-1)^p Q_{L_t}$ can have a non-negative limit as $t \to 0$ even when Q_{L_0} is not defined. This captures classes that are algebraic in the limit but not detected by any single polarisation.

Positivity under motivic convolution. For a product variety $X \times Y$, the Künneth decomposition gives $H^{p,p}(X \times Y) = \bigoplus_{\substack{a+c=p \\ b+d=p}} H^{a,b}(X) \otimes H^{c,d}(Y)$. For *rational* (p,p)-classes restricted to the diagonal summands $H^{a,a}(X) \otimes H^{p-a,p-a}(Y)$, the positivity condition on the product is multiplicative (Lemma V.10). The cross terms $H^{a,b}(X) \otimes H^{c,d}(Y)$ with $a \neq b$ contribute to the (p,p)-type only when the mixed Hodge types cancel, and their positivity analysis requires going beyond the diagonal framework of this paper.

Positivity after Galois averaging. For an absolutely Hodge class α, the Galois average $\bar{\alpha}$ (cf. Remark IX.22) is again an absolute Hodge class. For $p = 1$, $\bar{\alpha}$ is algebraic by the Lefschetz $(1,1)$-theorem. For $p \geq 2$, the algebraicity of $\bar{\alpha}$ is *not known* in general—it is equivalent to (a special case of) Deligne's question. The arithmetic positivity conditions AP1–AP3 can be reformulated as: α is algebraic if and only if α is "close" to its Galois average in the Q_L-metric. This reformulation is informative only if the Galois average itself can be shown to be algebraic independently of the Hodge conjecture.

Remark IX.22 (Galois averaging: profinite limit). The naïve averaging $\bar{\alpha} = \frac{1}{|\mathrm{Aut}(\mathbb{C}/\mathbb{Q})|} \sum_\sigma \sigma^*\alpha$ is ill-defined since $\mathrm{Aut}(\mathbb{C}/\mathbb{Q})$ is a profinite group of cardinality $2^{\mathfrak{c}}$. The correct formulation uses the projective limit: for each finite Galois extension $K/\mathbb{Q}$, form $\bar{\alpha}_K = \frac{1}{|\mathrm{Gal}(K/\mathbb{Q})|} \sum_{\sigma \in \mathrm{Gal}(K/\mathbb{Q})} \sigma^*\alpha$. The Galois average is then $\bar{\alpha} = \lim_K \bar{\alpha}_K$, which exists in $H^{2p}(X, \mathbb{Q})$ by the absoluteness of α.

H. Numerical illustration

Since $\mathrm{AP}^p = \mathrm{AbsHodge}^p$ is a *theorem* (not a conjecture), the following numerical computations serve as **pedagogical illustrations**, not as empirical verification. They demonstrate the AP framework concretely and illustrate the GHR spectrum on specific varieties.

The Python script `compute_ghr_spectrum.py` (accompanying this paper) computes the AP conditions across selected test cases:

- K3 surfaces (4 polarizations), abelian surfaces (5 polarizations), CM endomorphism classes on $E \times E$;
- $K3 \times K3$ tensor products with non-tensor-product perturbations.

All test configurations confirm $(-1)^p Q_L \geq 0$ on $P^{p,p}$, as guaranteed by the Hodge-Riemann relations. The $K3 \times K3$ test with non-tensor-product perturbations – classes that are *not* of type (p,p) after σ-conjugation, and hence not absolute Hodge – produces negative eigenvalues of $(-1)^p Q_L$, illustrating that the GHR spectrum can detect non-absoluteness. This does not go beyond Deligne's absoluteness criterion: the negative eigenvalues arise precisely because the perturbed classes fail to be (p,p)-classes on the conjugate variety, so the Hodge-Riemann relations do not apply.

I. Controlled test-family transfer beyond AP = AbsHodge

The post-Zookeeper interpretation of this paper is deliberately modest. Since arithmetic positivity collapses to absolute Hodge classes, the next step is not another global positivity axiom, but a test-family ladder for candidate strengthenings beyond Deligne's absoluteness. The symbols AP', AP'', and AP''' do not denote proposed global replacements for the Hodge conjecture. They denote increasingly restrictive tests on controlled families: AP' records provenance defects, AP'' asks for a bounded orbit-rigidity certificate, and AP''' filters out certificates whose auxiliary choices merely encode the target class. None of these conditions is claimed to imply the Hodge conjecture in general.

The current CORE closure of Riemann-$C2/MS2$ does not strengthen this paper's Hodge result. Hodge remains a negative control for the programme: Hodge–Riemann positivity saturates too broadly, so any future transfer must add genuinely new arithmetic or motivic constraints rather than reuse the Riemann kernel closure.

The five transfer slots are as follows.

Operator/form.: The Hodge Laplacian, Mumford–Tate and monodromy operators, and the period map.

Defect.: The residual transcendental part of an absolute Hodge class after subtracting the algebraic cycle span known in the family.

Approximation.: Abelian fourfolds of Weil type, singular OG6-type hyperkaehler models, powers of known examples, and reductions in characteristic p.

Gap.: Extra constraints beyond Hodge–Riemann positivity: p-adic slopes, Tate constraints, Mumford–Tate restrictions, and o-minimal algebraicity.

Limit.: Deformation, powers, and specialization from a controlled family to a broader motivic setting.

Accordingly, a useful next version should contain a table of test families. For each family it should record: known Hodge/Tate status, the cycle mechanism used in the literature, the proposed strengthened condition AP', and whether the defect vanishes. Weil fourfolds of discriminant one, OG6-type constructions, abelian varieties over finite fields satisfying the standard conjecture of Hodge type, and Voisin-type stress tests form the natural first ladder. If AP' still holds automatically on all negative controls, it is too weak; if it separates a known positive family from a stress case, it becomes a genuine candidate for further formalisation.

Towards AP′*: Provenance defects*

The first strengthening should be read as a provenance scanner rather than as another positivity cone. Relative to a named test family, define a defect vector

$$D_{\mathrm{prov}}(\alpha) = (D_{\mathrm{slope}}, D_{\mathrm{Tate}}, D_{\mathrm{MT}}, D_{\mathrm{mon}}, D_{\mathrm{cycle}}),$$

where the entries record, respectively, slope compatibility at good reductions, explanation by the Tate/Galois fixed part, Mumford–Tate compatibility, monodromy or o-minimal persistence, and the residual class left after subtracting the known algebraic cycle span. A conditional fingerprint statement would say that, in a test family where the relevant Tate/standard-conjecture input and comparison functoriality are independently available, $D_{\mathrm{prov}}(\alpha) = 0$ places α in the known algebraic cycle span. This is a special-family diagnostic, not a global theorem.

Towards AP″*: Conditional Orbit-Rigidity*

The five transfer slots suggest a prototype sharpening. Let $f\colon \mathcal{X}_S \to S$ be a spread-out of X over a smooth base S, let $Z \subset S$ be an irreducible algebraic component of the Hodge locus of α, let B denote the corresponding relative Chow or Hilbert component, and let $T \subset Z$ be a Zariski-dense subset at whose points α_t is algebraic relative to B. We call α *orbit-rigid on* (f, Z, B) if:

(i) *Properness.* The relative Chow/Hilbert components parametrising the relevant cycle classes are proper over S.

(ii) *Density.* T is Zariski-dense in Z.

(iii) *Comparison functoriality.* The comparison isomorphisms between Betti, de Rham, ℓ-adic, and crystalline realisations are functorial on the relevant quotient; no new residual classes arise when lifting from a special to the generic fibre.

(iv) *Specialisation rigidity.* Base change along $Z \to S$ does not create new residual cycle classes.

Lemma IX.23 (Conditional Orbit-Rigidity)**.** *Let $\alpha \in \mathrm{AbsHodge}^p(X)$ be orbit-rigid on (f, Z, B) in the sense above. Then the restriction α_η to the generic fibre $\mathcal{X}_\eta$ over the generic point η of Z lies in the image of the cycle class map $\mathrm{cl}\colon \mathrm{CH}^p(\mathcal{X}_\eta)_{\mathbb{Q}} \to H^{2p}(\mathcal{X}_\eta, \mathbb{Q}(p))$. If the base-point $0 \in S$ lies in the same controlled component, then α is algebraic.*

Proof idea. Properness of B *(i)* and Zariski-density of T *(ii)* imply via the valuative criterion that the closure of the algebraic locus in B contains the generic point η of Z. Comparison functoriality *(iii)* ensures that each algebraic class over $t \in T$ lifts consistently in all cohomological realisations. Specialisation rigidity *(iv)* prevents accumulation of transcendental residual classes at the limit. A complete proof requires the Cattani–Deligne–Kaplan principle for Hodge loci [12] together with a properness argument for the relative Hilbert scheme; both steps are conditional on the standard conjectures in the relevant characteristic-p reductions. □

Remark IX.24. Lemma IX.23 does not prove the Hodge Conjecture. It should be read as a conditional closure template, not as a new unconditional algebraicity theorem. Its force lies entirely in the independent verification of bounded relative Chow/Hilbert data, comparison functoriality, and specialisation rigidity in a named test family. Under those independently checked hypotheses, algebraicity propagates from a Zariski-dense set to the generic point. Weil abelian fourfolds of discriminant one are a natural first candidate for a positive test; $K3 \times K3$ perturbations outside the tensor-product locus serve as the first negative control. Whether AP″ separates these two families determines whether it constitutes a genuine strengthening beyond AbsHodge.

Towards AP‴*: Uniformity and anti-advice*

The orbit-rigidity certificate still has a failure mode: the family f, the Hodge-locus component Z, the Chow/Hilbert component B, or the dense set T could be chosen after seeing the target class α. In that case the certificate may encode α rather than explain it. We therefore use

$$\mathrm{AP}'''_{\mathrm{unif}}(\alpha; R, C)$$

as an anti-tautology filter for an AP″ certificate C. It requires that the auxiliary data are generated from a preregistered rule class R depending only on the ambient data (X, p) or on a fixed test-family ladder, with bounded advice, erosion stability under removal of an α-specific incidence condition, support on a neighbouring family or Zariski-dense set, and failure on negative controls driven only by Hodge–Riemann form stability. This condition does not add a new global Hodge conjecture; it says when a special-family certificate is non-tautological.

Family	**Cycle mechanism**	**AP′**	**AP″**	**AP‴**	**Status**
CM/Weil fourfolds	endomorphism cycles	slope/Tate/MT	candidate	rule from endomorphisms	positive control
Abelian fourfolds	known cycle span	Tate/MT	candidate	ambient family rule	positive control
OG6-type families	hyperkaehler cycles	monodromy/cycle	open	needs registry	test case
Char-p Tate cases	Tate classes	slope/Tate	conditional	reduction rule	arithmetic test
Voisin/$K3 \times K3$ stress	none fixed	residual class	should fail	post-hoc risk	negative control

Result classification

This paper is a **boundary result**: it determines precisely what positivity-based methods can and cannot achieve for the Hodge conjecture.

Result	Type
No-go theorem:	
AP = AbsHodge (HR adds nothing)	Theorem
HC reduces to Deligne's question	Corollary
Constructive tools:	
GHR spectrum definition	Definition
Finite pullback / primitive Künneth	Partial, conditional
Voisin-type GHR detection	Numerical
Conditional Orbit-Rigidity (AP$''$)	Lemma (conditional)
Uniform anti-advice filter (AP$'''$)	Methodological criterion
Open (equivalent to Deligne 1982):	
AbsHodge $\stackrel{?}{=}$ Algebraic	Open
Stronger positivity beyond HR	Research direction

X. CONCLUSION

We have proposed a reformulation of the Hodge Conjecture through the lens of positivity and No-Go obstructions. The key concept – *arithmetic positivity* – combines Hodge-Riemann positivity, absolute stability under $\mathrm{Aut}(\mathbb{C})$, and Lefschetz compatibility into a single condition.

A central finding of this paper (Remark IX.2) is that the arithmetically positive cone coincides with Deligne's absolute Hodge classes: $\mathrm{AP}^p(X) = \mathrm{AbsHodge}^p(X)$. This means the Arithmetic Positivity Conjecture is equivalent to Deligne's question "Are all absolute Hodge classes algebraic?" The positivity conditions (AP1), (AP2b), and (AP3) are automatically satisfied by any absolute Hodge class, via the Hodge-Riemann bilinear relations.

The paper does not introduce a genuinely new conjecture; the saturation of Hodge-Riemann positivity for (p,p)-classes is a standard consequence of the HR relations. The paper's contributions are expository and systematic:

1. **A new perspective:** The No-Go formulation recasts the Hodge Conjecture from a construction problem to an impossibility statement.
2. **A visualization of the absoluteness test:** The Galois-Hodge-Riemann spectrum reformulates Deligne's absoluteness criterion in the language of Hodge-Riemann positivity, making explicit which Galois conjugation could witness non-algebraicity. It does not provide constraints beyond Deligne's absoluteness.
3. **Partial functoriality checks:** The paper records partial functoriality checks rather than a full functorial theory: finite pullback is conditional on the ample-cone comparison, primitive Künneth products are controlled for product polarizations, and finite pushforward remains open.

The remaining challenge is Deligne's question: proving that absolute Hodge classes are algebraic. Future work should seek stronger positivity conditions – beyond the Hodge-Riemann relations – that could further narrow the gap between absolute Hodge classes and algebraic classes.

To summarize the philosophical position of this paper: the Hodge Conjecture, viewed through the positivity lens, does not ask us to build algebraic cycles – it asks us to show that geometry has no room for anything else. The present work shows that Hodge-Riemann positivity alone cannot accomplish this; genuinely new positivity conditions are required.

ACKNOWLEDGMENTS

This work was conducted as independent research without institutional funding.

AI tools. AI-based tools (Claude by Anthropic) were used for mathematical formalization and text generation. All mathematical content, conjectures, and responsibility for correctness lie solely with the author.

Funding information. This research received no external funding.

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